100

1. (10%) Order the following function by growth rate in increasing order:
n^2, $n\ log(log\ \mathrm{n})$, $n\ log(n^2)$, n^n

Ans: $n\,log(log\,n) < n\,log(n^2) < n^2 < n^n$

2. (10%) Ackerman's function $A(m,n)$ is defined as:

$$A(m,n) = \begin{cases} n+1, & \text{if } m = 0 \\ A(m-1,1), & \text{if } n = 0 \\ A(m-1, A(m, n-1)), & \text{otherwise} \end{cases}$$

What is the value of $A(3,4)$?

Ans: 125, my code is below

```
def A(m, n)->int:
    if m == 0:
        return n + 1
    if n == 0:
        return A(m - 1, 1)
    return A(m - 1, A(m, n - 1))
if __name__ == '__main__':
    print(A(3, 4))
```

問題 輸出 偵錯主控台 終端機 連接埠

[redacted]

125

3. (12%) Show that the following statements are correct:
(a) $5n^2 - 6n = \Theta(n^2)$
(b) $n! = O(n^n)$

Ans:

(a) Let $f(n) = 5n^2 - 6n$ and $g(n) = n^2$.
Take $c_1 = 1, c_2 = 6$ such that s.t. $\forall n \geq 1.5,\ c_1 g(n) \leq f(n) \leq c_2 g(n)$.
This shows that $5n^2 - 6n = \theta(n^2)$.

(b) Let $f(n) = n!$ and $g(n) = n^n$.
Take $c = 1$ such that s.t. $\forall n \geq 1,\ f(n) \leq cg(n)$.
This shows that $n! = O(n^n)$.

$n! = n(n-1)(n-2)\cdots 3\cdot 2\cdot 1 \leq \overbrace{n\cdot n\cdot\cdots n}^{n\text{个}}$ for $n \geq 1$

so $n! = O(n^n)$

4. (12%) Show that the following statements are incorrect:

(a) $10n^2 + 9 = O(n)$

(b) $3^n = O(2^n)$

Ans:

(a) Let $f(n) = 10n^2 + 9$ and $g(n) = n$.

There must exist a positive constant c such that $f(n) \leq cg(n), \forall n \geq n_0$

That means, $10n^2 + 9 \leq cn$

Take $c = 19$ such that $10n^2 + 9 \leq 19n$, this implies that $\frac{9}{10} \leq n_0 \leq 1$

It leads to contradiction, since n_0 contains an upper bound.

Therefore, the original statement is incorrect.

(b) Let $f(n) = 3^n$ and $g(n) = 2^n$

There must exist a positive constant c such that $f(n) \leq cg(n), \forall n \geq n_0 > 0$

That means, $3^n \leq c2^n$

Take $c = 1$ such that $3^n \leq 2^n$, this implies that $n \leq 0$

It leads to contradiction, since n_0 is not positive.

Therefore, the original statement is incorrect.

5. (16%) (from NTHU 106) Which ones of the following conjectures are correct? Please justify your answers:

(a) $3n^2 - 100n + 6 = O(n)$

(b) $log(n!) = \Theta(n \, log \, n)$

(c) $n! = O(n^n)$

(d) $(n+a)^b = \Theta(n^b)$ where a and b are real constants

(e) $(n+a)^b = \Omega(n^{b+1})$ where a and b are real constants

Ans:

(a) Incorrect, since $3n^2 - 100n + 6$ grow faster than cn

(b) Correct.

$log(n!) = log(1) + log(2) + \cdots + log(n) = \sum_{i=1}^{n} log(i) \leq nlog(n)$.

Thus, $log(n!) = O(nlogn)$. Let $f(n) = log(n!)$ and $g(n) = nlogn$

Take $c_1 = 0.5, c_2 = 1$ such that s.t. $\forall n \geq 1, c_1 g(n) \leq f(n) \leq c_2 g(n)$.

This shows that $log(n!) = \theta(nlogn)$.

(c) Correct. Let $f(n) = n!$ and $g(n) = n^n$.

Take $c = 1$ such that s.t. $\forall n \geq 1,\ f(n) \leq cg(n)$. This shows that $n! = O(n^n)$.

(d) Correct. Since $lim_{n\to\infty} \frac{(n+a)^b}{n^b} = 1$.

(e) Incorrect. Since $lim_{n\to\infty} \frac{(n+a)^b}{n^{b+1}} = 0$, this implies n^{b+1} grows faster.

6. (20%) Fill the step count table (yellow regions) for the following program:

Statement Let MAX_SIZE = n	s/e	Frequency	Total Steps
void *mult*(**int** *a*[][*MAX_SIZE*],**int** *b*[][*MAX_SIZE*], **int** *c*[][*MAX_SIZE*])	0	0	0
{	0	0	0
int *i*, *j*, *k*; n	0	0	0
for (*i* = 0; *i* < *MAX_SIZE*; *i*++) n	1	$n+1$	$n+1$
for (*j* = 0; *j* < *MAX_SIZE*; *j*++) {	1	$n(n+1)$	n^2+n
c [*i*][*j*] = 0;	1	n^2	n^2
for (*k* = 0; *k* < *MAX_SIZE*; *k*++)	1	$n^2(n+1)$	n^3+n^2
c [*i*][*j*] += *a* [*i*][*k*] * *b* [*k*][*j*];	1	n^3	n^3
}	0	0	0
}	0	0	0
Total			$2n^3+3n^2+2n+1$

7. (10%) Given an four-dimensional array A[100][200][300][400], if the address of A[3][20][32][45] is α + X, what is the value of X ?

Note: (1) α is the address of A[0][0][0][0]

(2) Only consider row major here.

Ans:

Using the formula: For array A[i][j][k][l], the address is $\alpha + i \cdot u_1 \cdot u_2 \cdot u_3 + j \cdot u_2 \cdot u_3 + k \cdot u_3 + l$, where u_0, u_1, u_2, u_3 are the maximum sizes of each dimension.

The answer is:

$X = 3 \cdot 200 \cdot 300 \cdot 400 + 20 \cdot 300 \cdot 400 + 32 \cdot 400 + 45 = 74412845$

8. (10%) (from NTCU 107) Consider the following function. Please derive the time complexity of function foo using recurrence relation. Please show your derivation step by step.

```c
int i[n];
int foo(int a, int b, int c) {
      int d;
      if (a>b) return -1;
      d=(a+b)/2;
      if (c==i[d])
            return d;
      else if (c>i[d])
            return foo(d+1,b,c);
      else
            return foo(a, d-1,c);
}
```

Statement	s/e	Freq	Total
`int foo(int a, int b, int c) {`	0	0	0
`int d;`	0	0	0
`if (a>b) return -1;`	1	t	t
`d=(a+b)/2;`	1	t	t
`if (c==i[d])`	1	t	t
`return d;`	1	1	1
`else if (c>i[d])`	1	t	t
`return foo(d+1,b,c);`	1	1	1
`else`	1	t	t
`return foo(a, d-1,c);`	1	~~1~~ t	~~1~~ t
`}`			5t+3

Ans:

Let recursion runs t times

In the worst case, this function would cut array i[n] into half and half by parameters a and b until $n = 2^t$, which means the array can't cut in half anymore.

Thus, $t = log_2 n$

So, the time complexity is $5log_2 n + 3 = O(logn)$